**Semester One
Examination 2017
Question/Answer booklet**

# MATHEMATICS APPLICATIONS UNIT 1

## Section Two: Calculator-assumed

Student Name: ____________________________________

Teacher's Name: ____________________________________

## Time allowed for this section

Reading time before commencing work: ten minutes
Working time for paper: one hundred minutes

## Material required/recommended for this section

**To be provided by the supervisor**
This Question/Answer booklet
Formula Sheet (retained from Section One)

**To be provided by the candidate**
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction tape/fluid, erasers, ruler, highlighters

Special Items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in the WACE examinations.

## Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

## Structure of this paper

| | Number of questions available | Number of questions to be attempted | Working time (minutes) | Marks available | Percentage of exam |
|---|---|---|---|---|---|
| Section One Calculator—free | **13** | **13** | **50** | **50** | **35** |
| **Section Two Calculator—assumed** | 15 | 15 | 100 | 100 | 65 |
| | | | | | 100 |

## Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2017.* Sitting this examination implies that you agree to abide by these rules.

2. Answer the questions according to the following instructions.

   **Show all your working clearly.** Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat an answer to any question, ensure that you cancel the answer you do not wish to have marked.

   It is recommended that you **do not use pencil**, except in diagrams.

3. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
   - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.

5. The Formula Sheet is **not** handed in with your Question/Answer Booklet.

**See next page**

## Section Two: Calculator–assumed 65% (100 marks)

This section has **fifteen (15)** questions. Attempt **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Working time: 100 minutes

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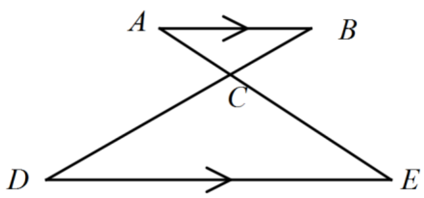
**Question 14 (5 marks)**

Two triangles are shown below. AB is parallel to DE.




(a) Give reasons why triangle ABC is similar to triangle EDC. (2 marks)

AB : ED = 1 : 2

(b) If AC has length of 4 units, determine the length of AE. (3 marks)

**Question 15 (8 marks)**

Pipes-R-Us claim that customers ordering lengths of pipe always receive at least 8% more than they order.

(a) Assuming the claim is true, determine the minimum length of pipe that Tony receives if he purchases 6 metres of pipe. (2 marks)

Dianne purchases 12 metres of pipe and receives 13 metres.

(b) Is the claim correct in Dianne's case? Show calculations to support your answer. (3 marks)

Pipe costs \$3 per metre. Martha received 6 metres.

(c) What is the greatest amount that Martha paid for her order? Assume the claim was correct. (3 marks)

**Question 16** **(8 marks)**

Mildred inherits \$25 000. She decides to invest all of the money for 1 year while she considers what to do with it.

Her investment options are :
Bank A. Interest rate is 5% pa compounded annually.
Bank B. Interest rate is 4.5% pa compounded monthly.
Bank C. Interest rate is 4.8% pa compounded every 6 months.

(a) Calculate the interest earned in each situation and hence determine which of the options will result in Mildred earning the most interest. (4 marks)

Mildred believes that compound interest always provides more interest than simple interest on the same principal, at the same rate and for the same time.

(b) Is that true if Bank A is compared with a simple interest option? Give calculations to explain your answer. (2 marks)

Gertrude wishes to invest some money to earn \$10 000 in interest each year.

(c) How much money must Gertrude invest in Bank B to achieve that goal? (2 marks)

**Question 17 (9 marks)**

The tax table for Australia for 2016 –17 is shown below.

| Taxable Income | Tax on this income |
|---|---|
| 0 – $18 200 | Nil |
| $18 201 – $37 000 | 19c for each dollar over $18 200 |
| $37 001 – $80 000 | $3 572 plus 32.5c for each $1 over $37 000 |
| $80 001 – $180 000 | $17 547 plus 37c for each $1 over $80 000 |
| $180 001 and over | $54 547 plus 45c for each $1 over $180 000 |

(a) Mary paid $17 547 in tax. What was her taxable income? (1 mark)

(b) (i) Carolyn earns $120 000. She has deductions of $20 000.
Show that Carolyn should have paid $24 947 during that financial year. (2 marks)

(ii) Carolyn actually paid $1 000 per fortnight in tax from her pay.
How much will she have refunded when she submits her tax return? (2 marks)

Politician, Ronald Rump suggests that to be fair to everyone, the tax system should change.
He wants everyone to pay a flat tax of $x$% of their taxable income.
Carolyn calculates that her tax payable would be halved under that system.

(iii) Determine $x$. (4 marks)

**Question 18** **(6 marks)**

The following table shows the pay rates at Trucks – R – Us.

| Day | Rate per hour |
|---|---|
| Monday-Friday | $25 |
| Saturday | $x$ |
| Sunday | Double Time |

Last week, Dan worked the following hours:

| Day | Hours worked |
|---|---|
| Monday-Friday | 30 |
| Saturday | 10 |
| Sunday | $y$ |

(a) Calculate the amount Dan earned for his Monday – Friday work. (1 mark)

Dan earned $375 for his Saturday work.

(b) What was Dan's rate of pay on the Saturday? (1 mark)

Dan earned $400 for his Sunday work.

(c) How many hours did Dan work last Sunday? (1 mark)

Dan's employer deposited an amount equal to 9.5% of Dan's pay into Dan's superannuation account.

(d) (i) Briefly describe the purpose of superannuation. (1 mark)

(ii) How much money was deposited into Dan's superannuation account for last week? (2 marks)

**Question 19** **(7 marks)**

A map has a scale of 1:40 000 [ie 1 cm on the map represents 40 000 cm (400 m) in real life.]

(a) What length is represented by 5 cm on the map? (1 mark)

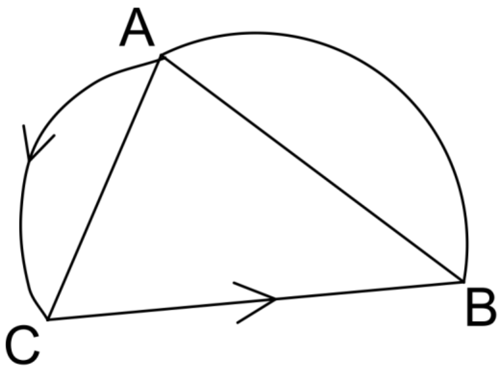
(b) How many mm on the map represent 1 km in real life? (2 marks)

A square region on the map has an area of 4 $cm^2$.

(c) Find the length of the diagonal drawn on the real life region. (4 marks)

**Question 20 (6 marks)**

The network below shows possible paths (2 way and 1 way) between vertices A, B and C.




(a) Complete the matrix, X, which represents the one stage routes shown on the network.
(2 marks)

| | | To | | |
|---|---|---|---|---|
| From | | A | B | C |
| | A | 0 | 2 | 2 |
| | B | | | |
| | C | | | |

(b) Complete matrix $X^2$ and explain the meaning of the entry (6) in Row A Column A.
(4 marks)

| | | To | | |
|---|---|---|---|---|
| From | | A | B | C |
| | A | 6 | | |
| | B | | | |
| | C | | | |

**Question 21** **(3 marks)**

Matrix A is 4 by 7. Matrix B has $x$ rows and $y$ columns.

(a) Determine $x$ and $y$ if A + B exists. (1 mark)

(b) Determine $x$ and $y$ if AB has 4 rows and 5 columns. (1 mark)

(c) Determine $x$ and $y$ if BA has 3 rows. (1 mark)

**Question 22** **(4 marks)**

A triangle, $T_1$, has sides of length 7, 8 and 9 cm.
A triangle, $T_2$, similar to $T_1$, has a perimeter of 72 cm.

(a) Determine the lengths of the sides of $T_2$. (3 marks)

It can be shown that the area of $T_1$ = $\sqrt{720}$ cm$^2$

(b) Determine the area of $T_2$. (1 mark)

**Question 23** **(8 marks)**

Houses in Leem Creek were appreciating in value at an annual rate of 6%.
An average home in that suburb was valued at $550 000.

(a) Determine the price of that average home 5 years after the valuation. (2 marks)

After reaching an average value of $700 000 at the beginning of 2010, values began to drop by 4% pa.

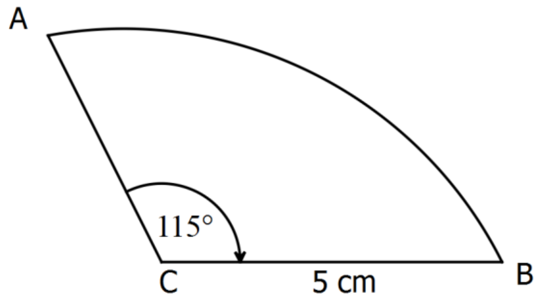
(b) Determine the average value in the suburb at the beginning of 2017. State your answer to a sensible amount of accuracy. (2 marks)

(c) (i) Determine during which year the value will drop to $500 000. (3 marks)

(ii) Give a reason why that drop to a value of $500 000 is unlikely. (1 mark)

**Question 24 (5 marks)**

A student used a protractor of radius 5 cm to mark an angle of 115 degrees on his paper. (3 marks)




(a) Determine the arc length from A to B. Use 2 decimal places in your answer. (2 marks)

(b) Determine the area of sector ABC. (2 marks)

A teacher had a protractor with a radius of 50 cm.
She drew the same shape, with an angle of 115°, on the white board.

(c) Determine the area of the whiteboarded sector of the same shape. (1 mark)

**Question 25 (6 marks)**

Frances was holidaying in Bali for a week.
She exchanged $500 Australian for Indonesian rupiah. Each dollar was worth 10 000 rupiah.

(a) How many rupiah would Frances receive for $500 AUD? (1 mark)

Frances saw a beautiful dress which cost 283 000 rupiah.

(b) What was the price in Australian dollars?

(2 marks)

Vegemite was available in Bali. There were 2 sizes of vegemite jars.
300 g jars cost 21 000 rupiah.
750 g jars cost 48 000 rupiah.

(c) (i) Based on price, which is the better buy? Show some supporting calculation. (2 marks)

(ii) State a sensible reason why Gertrude chose the more expensive option. (1 mark)

**Question 26 (13 marks)**

On leaving school, Dan purchases a \$15 000 car on credit. He makes a deposit and borrows the remainder from the bank. His repayment spreadsheet is shown below.
Dan makes repayments at the end of every month.

| Start of month | Opening Balance | Interest for the month | Repayment | Closing Balance |
|---|---|---|---|---|
| 1 | 12 000 | 120 | 400 | *x* |
| 2 | 11720 | 117.20 | 400 | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| 36 | 335.12 | 3.35 | *y* | 0 |

(a) Calculate the annual advertised interest rate using the information for month 1. (2 marks)

(b) Show that the rate of interest charged did not change throughout the loan period. (2 marks)

(c) How many years did the loan take to repay? (1 mark)

(d) Calculate the value of $x$. (1 mark)

(e) Calculate the value of $y$. (1 mark)

(f) How much did he pay in total for the car? (3 marks)

(g) How much interest did he pay to the bank? (2 marks)

A rival bank states "You can save up to or over $1 000 by buying through us".

(h) Mathematically, what is wrong with this statement? (1 mark)

**Question 27 (6 marks)**

The radius of a sphere is 3 cm.

(a) Show that the volume of the sphere is numerically the same as the surface area of the sphere.
(3 marks)

(b) Will the situation in (a) be the case for other sized spheres? Explain with calculations.
(3 marks)

**Question 28 (6 marks)**

Determine the ratio of surface area for figure 1 compared with surface area of figure 2. Units are cm.
(6 marks)

Figure 1

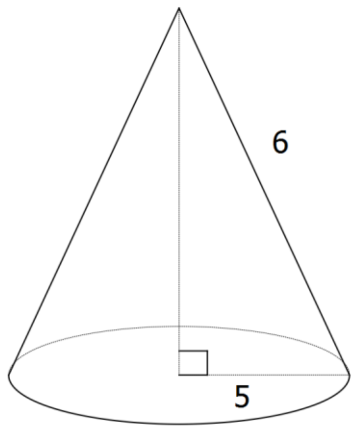
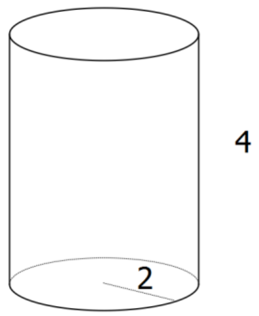



Figure 2




**End of questions**

**Additional working space**

Question number(s): ……………………

**Additional working space**

Question number(s): ……………………

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